

MS Algebra Recitation 3

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1 Good Group Actions to Know 2: Electric Boogaloo

Definition. Let $G \curvearrowright G$ by conjugation (i.e. $g \cdot h = ghg^{-1}$). The *centralizer* of $h \in G$ is the stabilizer of h under this action. That is,

$$Z_G(h) = G_h = \{g \in G : ghg^{-1} = h\} = \{g \in G : gh = hg\}.$$

Definition. Let $G \curvearrowright G$ by conjugation. The *center* of G is the kernel of the map $G \rightarrow S_G$. That is,

$$Z(G) = \{g \in G : ghg^{-1} = h \ \forall h \in G\} = \{g \in G : gh = hg \ \forall h \in G\}.$$

Definition. Let $G \curvearrowright \{H : H \leq G\}$ by conjugation (verify that this is a well-defined action!). The *normalizer* of H is the stabilizer of H under this action. That is,

$$Z_G(H) = G_H = \{g \in G : gHg^{-1} = H\}.$$

The *centralizer* of H is the pointwise normalizer of H :

$$Z_G(H) = \{g \in G : ghg^{-1} = h \ \forall h \in H\} = \bigcap_{h \in H} Z_G(h).$$

Because of the way we defined these, we immediately get the following:

- Centralizers are subgroups
- Normalizers are subgroups
- The center is a *normal* subgroup

Also straightforward from the definition is that $H \trianglelefteq N_G(H)$ for any $H \leq G$. This ends up being pretty useful, since even if we don't have a normal subgroup, we can restrict and assume our subgroup is normal.

The following are exercises in symbol pushing:

- $g \in Z_G(h)$ if and only if $h \in Z_G(g)$
- $Z(G)$ is the set of fixed points of the action $G \curvearrowright G$ by conjugation
- $Z_G(g)$ is the set of fixed points of $\varphi_g : h \mapsto ghg^{-1}$
- $Z(G) = \bigcap_{g \in G} Z_G(g)$

2 Why Do We Care About Normal Subgroups?

We first give an easy way to generate normal subgroups.

Proposition. Let G be a group and $\phi : G \rightarrow H$ be a homomorphism. Then $\ker \phi \trianglelefteq G$.

Proof. Recall that $K \trianglelefteq G$ if and only if $gKg^{-1} \subseteq K$ for all $g \in G$. Set $K = \ker \phi$ and take $k \in K$ and $g \in G$. Then,

$$\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g)^{-1} = \phi(g)1_H\phi(g)^{-1} = \phi(g)\phi(g)^{-1} = 1_H.$$

So $gKg^{-1} \subseteq K$ and $K \trianglelefteq G$. ■

To motivate the definition of a normal subgroup, we discuss how one would want to construct a quotient group. Suppose G is a group and let $\sim$ be an equivalence class on G . Then we want to make $G/\sim$ into a group via $[g][h] = [gh]$. For this to be well-defined, it needs to be equivalent regardless of what representative we pick for each equivalence class. That is, if $g \sim g'$ and $h \sim h'$, then we need $gg' \sim hh'$. We'll consider a slight weakening of this property.

Proposition. *Let G be a group and $\sim$ be an equivalence relation on G with $g \sim g' \implies hg \sim hg'$ for all $g, g', h \in G$. Then $H := [1]$ is a subgroup of G , and $[g] = gH$.*

Proof. We first verify that $H \leq G$. Clearly $1 \in H$ so H is nonempty. If $g, g' \in H$ then we have $1 \sim g$ and $1 \sim g'$. Multiplying $1 \sim g'$ on the left by g gives $g \sim gg'$. Transitivity then gives $1 \sim g \sim gg'$. Similarly, multiplying $1 \sim g$ on the left by g^{-1} gives $g^{-1} \sim 1$, and reflexivity gives $1 \sim g^{-1}$. Thus, $H \leq G$.

Now, if $g \sim g'$ then $1 \sim g^{-1}g'$ and so $g^{-1}g' \in H$. Thus, $gH = g'H$ and so the equivalence class of g is contained in gH . Conversely, if $gH = g'H$ then $g^{-1}g' \in H$, so $g^{-1}g' \sim 1$ and $g' \sim g$. Thus, $[g] = gH$. ■

Symmetrically, if $\sim$ satisfies $g \sim g' \implies gh \sim g'h$, then the equivalence classes are the right cosets of $[1]$.

Exercise. Show that $\sim$ satisfies

$$g \sim g' \implies hg \sim hg', \quad g \sim g' \implies gh \sim g'h$$

if and only if it satisfies

$$g \sim g', h \sim h' \implies gh \sim g'h'.$$

So, to have a well-defined group structure on $G/\sim$, the left cosets of $[1]$ must be equal to the right cosets of $[1]$. That is, $[1]$ must be a normal subgroup. Tracing through all the previous proofs also shows that, for a normal $N \trianglelefteq G$, quotienting by the equivalence relation $g \sim g' \iff gN = g'N$ can yield a well-defined group structure.

Definition. Let G be a group and $N \trianglelefteq G$ be a normal subgroup. The *quotient group* G/N is the set of left cosets of N under the operation $(gN)(g'N) = (gg')N$.

Remark. Let's show "by hand" that this group operation is well-defined. Suppose $gN = g'N$ and $hN = h'N$. We want to show that $ghN = g'h'N$. We have that $g^{-1}g', h^{-1}h' \in N$. Now,

$$(gh)^{-1}(g'h') = h^{-1}(g^{-1}g')h' = h^{-1}nh',$$

where we write $n = g^{-1}g' \in N$. Since N is normal, we can find $n' \in N$ with $nh' = h'n'$ (convince yourself that this follows from $(h')^{-1}Nh' \subseteq N$). Thus,

$$h^{-1}nh' = h^{-1}h'n' = (h^{-1}h')n' \in N,$$

as desired.

Exercise. We've shown that G/N has a well-defined binary operation. Show that it is indeed a group with identity $1N$ and inverses $(gN)^{-1} = g^{-1}N$.

So why do we care about quotient groups? I'll give you two rough reasons. The first is the following theorem.

Theorem (First Isomorphism Theorem). *Let G be a group and $\phi : G \rightarrow H$ a homomorphism. Then $G/\ker \phi \cong \text{im } \phi$.*

So quotient groups give us an intrinsic way of describing how homomorphic images of G can look. It also in particular tells us that there are only so many ways that such images can look.

Here's our second:

Theorem (Fourth Isomorphism Theorem). *Let G be a group and $N \trianglelefteq G$. Then there is an order-preserving bijection between subgroups of G containing N and subgroups of G/N .*

That is, the "subgroup structure" of G/N appears in the "subgroup structure" of G . This is then useful because studying subgroups of G/N tells us something about G . It also tends to be useful because we can then do things like induct on $|G|$ and use the inductive hypothesis on G/N .

3 The 5/8ths Theorem

Let's go through an example of how to work with normal subgroups.

Proposition. *Let G be a group. If $G/Z(G)$ is cyclic then G is abelian.*

Proof. Write $Z = Z(G)$. If G/Z is cyclic, then it is generated by one element, say $G/Z = \langle gZ \rangle$. Take $h, k \in G$. Then $hZ \in \langle gZ \rangle$, so there is some $n \in \mathbb{Z}$ with $hZ = (gZ)^n = g^n Z$. This means we can find some $z_1 \in Z$ with $h = g^n z_1$. Symmetrically, there are $m \in \mathbb{Z}$ and $z_2 \in Z$ with $k = g^m z_2$.

Now, since $z_1, z_2 \in Z$, they commute with all elements of G . So, we have that

$$hk = (g^n z_1)(g^m z_2) = g^n (z_1 g^m) z_2 \stackrel{z_1 \in Z}{=} g^n g^m z_1 z_2.$$

Since $g^n g^m = g^{n+m} = g^m g^n$, we have that this is equal to

$$g^m g^n z_1 z_2 \stackrel{z_2 \in Z}{=} g^m z_2 g^n z_1 = kh.$$

That is, $hk = kh$, so G is abelian. ■

In particular, since we know that the only group of order p for p a prime is the cyclic group C_p , we see that $|G/Z(G)| = [G : Z(G)] \neq p$ for all primes p whenever G is nonabelian. Since $G \neq Z(G)$ for nonabelian G , this also gives us the following corollary:

Corollary. *If G is nonabelian, then $[G : Z(G)] \geq 4$.*

We'll use this as a lemma in the following.

Theorem. *Let G be a finite group. If the probability that two randomly chosen elements of G commute is $> 5/8$, then G is abelian.*

Proof. Let G be nonabelian. Let x and y be randomly chosen elements of G . Then,

$$\mathbb{P}(x, y \text{ commute}) = \mathbb{P}(x, y \text{ comm} | x \in Z(G)) \cdot \mathbb{P}(x \in Z(G)) + \mathbb{P}(x, y \text{ comm} | x \notin Z(G)) \cdot \mathbb{P}(x \notin Z(G)).$$

Set $p = \mathbb{P}(x \in Z(G))$. Since elements of $Z(G)$ by definition commute with all other elements of G , we have that

$$\mathbb{P}(x, y \text{ comm}) = 1 \cdot p + \mathbb{P}(x, y \text{ comm} | x \notin Z(G)) \cdot (1 - p).$$

Now, x and y commuting is equivalent to $y \in Z_G(x)$. If $x \notin Z(G)$, then $Z_G(x) \neq G$, and so $[G : Z_G(x)] \geq 2$ and $\mathbb{P}(y \in Z_G(x)) = \frac{|Z_G(x)|}{|G|} = \frac{1}{[G : Z_G(x)]} \leq \frac{1}{2}$. Thus,

$$\mathbb{P}(x, y \text{ comm}) = p + \mathbb{P}(y \in Z_G(x) | x \notin Z(G)) \cdot (1 - p) \leq p + \frac{1}{2}(1 - p) = \frac{1}{2} + \frac{1}{2}p.$$

Now, $p = \mathbb{P}(x \in Z(G)) = \frac{|Z(G)|}{|G|} = \frac{1}{[G : Z(G)]}$, which is $\leq \frac{1}{4}$ by the previous corollary. Thus,

$$\mathbb{P}(x, y \text{ comm}) = \frac{1}{2} + \frac{1}{2}p \leq \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{5}{8}.$$

This bound is sharp: the group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with multiplication $i^2 = j^2 = k^2 = ijk = -1$ is not abelian but the above probability is exactly $5/8$. ■

4 Using Orbit-Stabilizer to Count

Proposition. *Let G be a finite group and $H < G$ be a proper subgroup. Then there exists some $k \in G$ with $k \notin \bigcup_{g \in G} gHg^{-1}$. That is, there is some element in G that is not conjugate to any element of H .*

Proof. We'll first present the naive way of approaching this. We consider $G \curvearrowright \mathcal{P}(G)$ via conjugation. The sets gHg^{-1} are then the elements of the orbit of H . Since $|gHg^{-1}| = |H|$ (check this!), we have an upper bound

$$\left| \bigcup_{g \in G} gHg^{-1} \right| \leq |H| \cdot |\mathcal{O}_H|.$$

Now, $|\mathcal{O}_H| = [G : G_H]$ by orbit stabilizer. In this context, $G_H = N_G(H)$ is the normalizer, and so

$$\left| \bigcup_{g \in G} gHg^{-1} \right| \leq |H| \cdot [G : N_G(H)] = |G| \cdot \frac{|H|}{|N_G(H)|}.$$

If $|H|/|N_G(H)| < 1$, this shows that we must be able to find some $g \in G$ not conjugate to any element of H . Unfortunately, this isn't always true.

To salvage this proof, we note that we were being overly conservative in our previous bound. We assumed that different conjugates of H are disjoint. However, this isn't true: $1 \in gHg^{-1}$ for all $g \in G$. So, we'll split H into $\{1\}$ and $H \setminus \{1\}$. This gives us the sharper bound

$$\left| \bigcup_{g \in G} gHg^{-1} \right| = 1 + (|H| - 1) \cdot |\mathcal{O}_H| = |G| \cdot \frac{|H|}{|N_G(H)|} - ([G : N_G(H)] - 1).$$

Just as before, if $|H|/|N_G(H)| < 1$ then this is $< |G|$ and we are done. Suppose that $|H| = |N_G(H)|$. Since $H \leq N_G(H)$, this gives us that $N_G(H) = H < G$ is a proper subgroup of G . Thus, $[G : N_G(H)] \geq 2$, and so $[G : N_G(H)] - 1 \geq 1$. Thus, in either case we see that $\left| \bigcup_{g \in G} gHg^{-1} \right| < |G|$ and so there is some $k \in G$ not conjugate to any element of H . ■